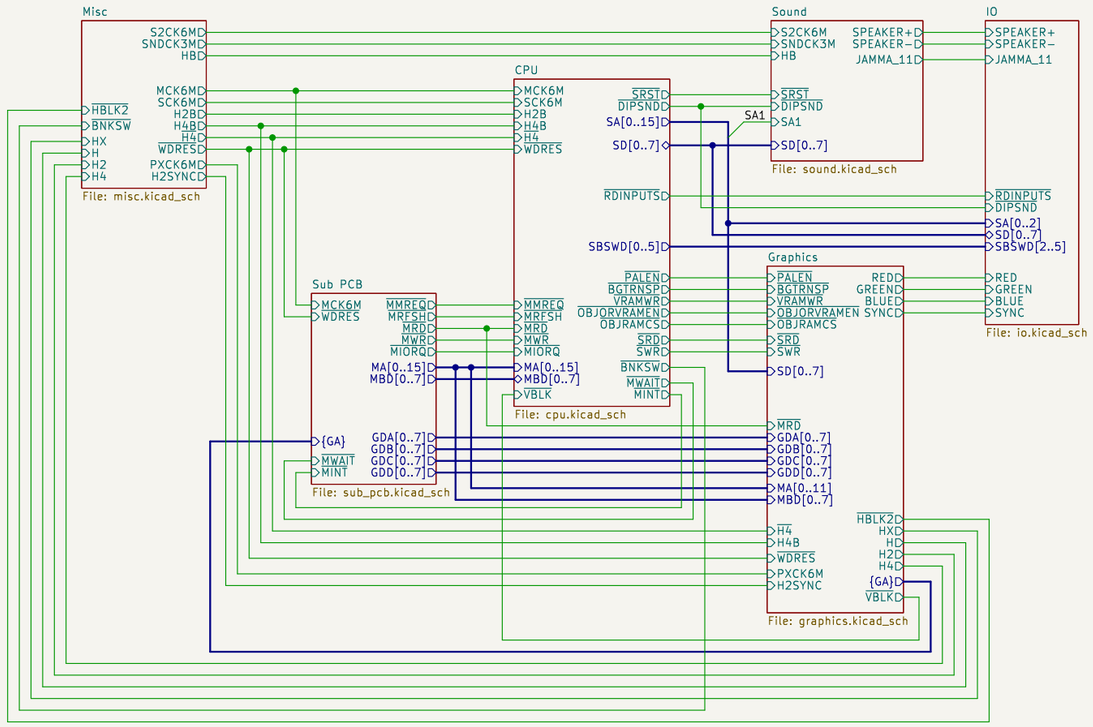


Note: The reference designator refers to the position on the PCB. If the reference designator already exists the letter is doubled. I.e. C1 -> CC1.

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Sheet: /		
File: tnzs.kicad_sch		
Title: The NewZealand Story – P0-043A		
Size: A3	Date: 2024-04-07	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/7

IC5
74LS174
MD0 MD1 MD2 MD3 MD4 MD5
D0 D1 D2 D3 D4 D5
Q0 Q1 Q2 Q3 Q4 Q5
Cp
hex flip flop
Mr
VCC

IC1
74F245
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
A0 A1 A2 A3 A4 A5 A6 A7
B0 B1 B2 B3 B4 B5 B6 B7
buffers
A->B
MRD
CE
MBD0 MBD1 MBD2 MBD3 MBD4 MBD5 MBD6 MBD7
MBD[0..7]

MA11 MA12 MA13 MA14 MA15
MCK6M
MBD4 MBD3 MBD5 MBD6 MBD2 MBD7 MBD0 MBD1
MINT MBUSRQ MHALT MMREQ MIORQ

JS2
Conn_01x20
VCC GND

JS1
Conn_01x20
MA10 MA9 MA8 MA7 MA6 MA5 MA4 MA3 MA2 MA1 MA0
MRFSH MM1 WDRES MBUSRQ MWAIT MBUSACK MWR MRD

NC: MM1, MBUSRQ, MHALT, MBUSACK

IC6
PAL16L8'
MA10 MA11 MA12 MA13 MA14 MA15 GROMEN MWR MRD MMREQ
I1 I2 I3 I4 I5 I6 I7 I8 I9 I11
I/O13 I/O14 I/O15 I/O16 I/O17 I/O18 O12 O19
GROMENB MDC2 MDC1 MDC0 MA9 MRAMCS
VCC VSS GND

/MRAMCS = /MA14 & MA15 & /MDC2 & /MDC1
o13 = /MA14 & MA15 & /MDC2 & /MDC1
/GROMENB = GROMEN
/o19 = MA9 & MA10 & /MA11 & MA12 & MA14 & MA15 & /MWR & /MMREQ

MA[0..15]

IC3
Z80CPU
WDRES MINT MM1 MRFSH MWAIT MHALT MRD MWR MMREQ MIORQ MBUSRQ MBUSACK
RESET NMI INT M1 RFSH WAIT HALT RD WR MREQ IORQ BUSRQ BUSACK
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15
MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MA13 MA15
D0 D1 D2 D3 D4 D5 D6 D7
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
VCC GND

IC4C
74HC08
WDRES MCK6M VCC MINT

IC2
SRM20256
MA0 MA1 MA2 MA3 MA4 MA5 MA6 MA7 MA8 MA9 MA10 MA11 MA12 MA13 MDC0
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14
Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7
MD0 MD1 MD2 MD3 MD4 MD5 MD6 MD7
CS OE WE
MRD MWR
VCC GND

MD[0..7]

IC7 IC9 IC11 IC12
IC8 IC10 IC13 IC14
27C010
GA0 GA1 GA2 H8R GA4 GA5 GA6 GA7 GA8 GA9 GA10 GA11 GA12 GA13 GA14 GA15 GA16
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 A15 A16
D0 D1 D2 D3 D4 D5 D6 D7
GDA0 GDA1 GDA2 GDA3 GDA4 GDA5 GDA6 GDA7
GDB0 GDB1 GDB2 GDB3 GDB4 GDB5 GDB6 GDB7
GDC0 GDC1 GDC2 GDC3 GDC4 GDC5 GDC6 GDC7
GDD0 GDD1 GDD2 GDD3 GDD4 GDD5 GDD6 GDD7
PGM VPP CE OE
VCC GND
GROMEN
GROMENB

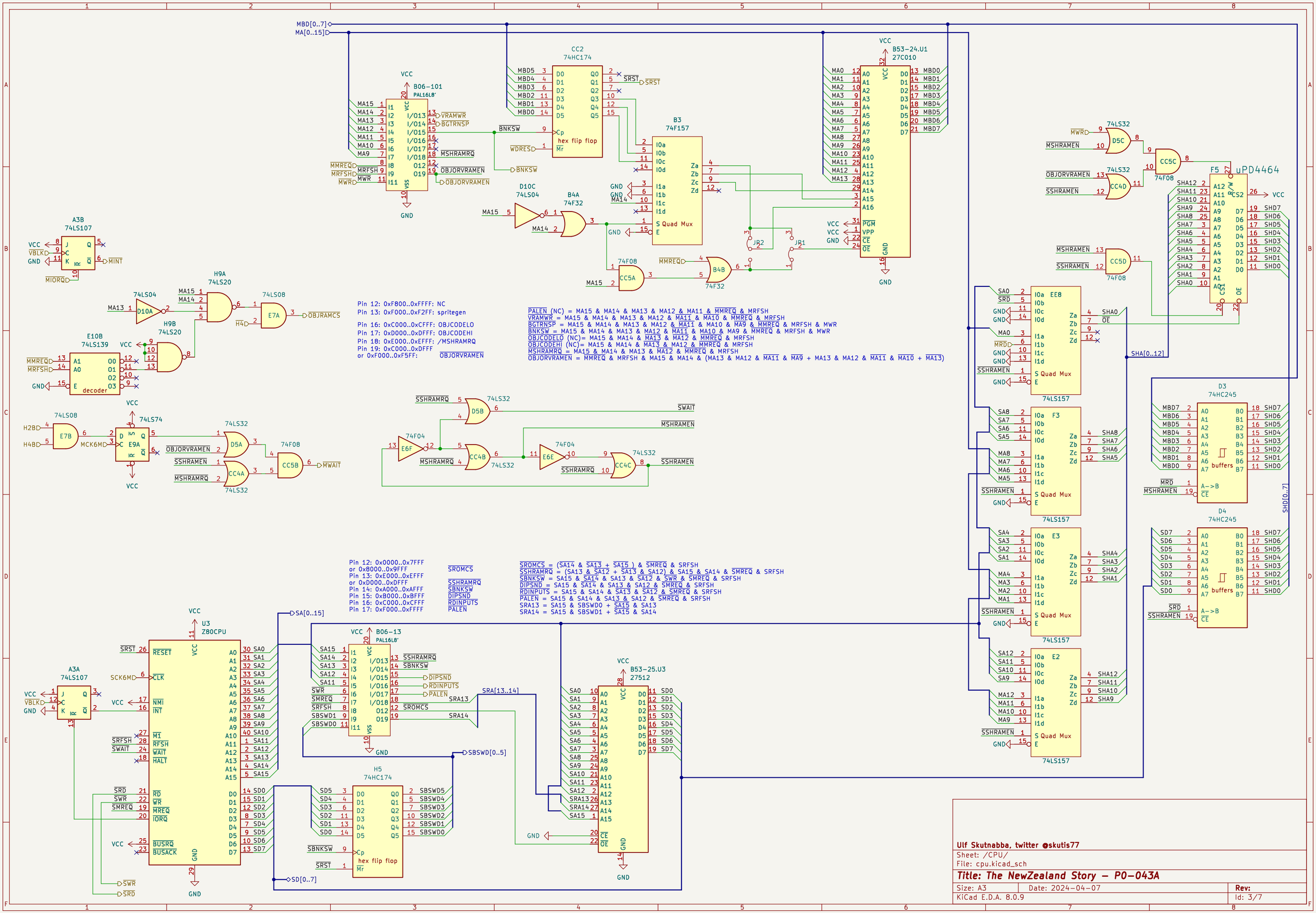
{GA}

JS6 JS5 JS4 JS3
JS10 JS9 JS8 JS7
Conn_01x20
GND VCC
GDB0 GDB1 GDB2 GDB3 GDB4 GDB5 GDB6 GDB7
GDA0 GDA1 GDA2 GDA3 GDA4 GDA5 GDA6 GDA7
GA8 GA7 GA6 GA5 GA4 H8R GA2 GA1 GA9 GA10 GA11 GA12 GA13 GA14 GA15 GA16 GROMEN GA0
GDD0 GDD1 GDD2 GDD3 GDD4 GDD5 GDD6 GDD7
GDC0 GDC1 GDC2 GDC3 GDC4 GDC5 GDC6 GDC7

GDA[0..7]
GDB[0..7]
GDD[0..7]
GDC[0..7]

IC4D
74HC08
IC4B
74HC08
IC4A
74HC08
GND

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Sheet: /Sub PCB/
File: sub_pcb.kicad_sch
Title: The NewZealand Story – PO-043A
Size: A3 | Date: 2024-04-07 | Rev:
KiCad E.D.A. 8.0.9 | Id: 2/7


MBD[0..7]
MA[0..15]
CC2
74HC174
B06-101
PAL16L8'
B3
74F157
B53-24.U1
27C010
VCC
GND
MMREQ
MRFSH
MWR
VRAMWR
BGTRNSP
MSHRAMRQ
OBJORVRAMEN
BNKSW
WDRES
SRST
D10C
74LS04
B4A
74F32
CC5A
74F08
B4B
JP2
JP1
A3B
74LS107
MINT
MIORQ
H9A
74LS20
H9B
E10B
74LS139
decoder
E7A
OBJRAMCS
H4D
Pin 12: 0xF800..0xFFFF: NC
Pin 13: 0xF000..0xF2FF: spritegen
Pin 16: 0xC000..0xCFFF: OBJCODELO
Pin 17: 0xD000..0xDFFF: OBJCODEHI
Pin 18: 0xE000..0xEFFF: /MSHRAMRQ
Pin 19: 0xC000..0xDFFF
or 0xF000..0xF5FF: OBJORVRAMEN
PALEN (NC) = MA15 & MA14 & MA13 & MA12 & MA11 & MMREQ & MRFSH
VRAMWR = MA15 & MA14 & MA13 & MA12 & MA11 & MA10 & MMREQ & MRFSH
BGTRNSP = MA15 & MA14 & MA13 & MA12 & MA11 & MA10 & MA9 & MMREQ & MRFSH & MWR
BNKSW = MA15 & MA14 & MA13 & MA12 & MA11 & MA10 & MA9 & MMREQ & MRFSH & MWR
OBJCODELO (NC)= MA15 & MA14 & MA13 & MA12 & MMREQ & MRFSH
OBJCODEHI (NC)= MA15 & MA14 & MA13 & MA12 & MMREQ & MRFSH
MSHRAMRQ = MA15 & MA14 & MA13 & MA12 & MMREQ & MRFSH
OBJORVRAMEN = MMREQ & MRFSH & MA15 & MA14 & (MA13 & MA12 & MA11 & MA9 + MA13 & MA12 & MA11 & MA10 + MA13)
74LS08
74LS74
74LS32
D5A
D5B
D5C
CC4A
CC4B
CC5B
CC5C
CC5D
74F04
E6F
E6E
E7B
E9A
SSHRAMRQ
MCK6MD
H2B
H4B
SWAIT
MWAIT
MSHRAMEN
SSHRAMEN
OBJORVRAMEN
MSHRAMRQ
F5
uPD4464
R/W
CS1
CS2
OE
SHA12
SHA11
SHA10
SHA9
SHA8
SHA7
SHA6
SHA5
SHA4
SHA3
SHA2
SHA1
SHA0
SHD7
SHD6
SHD5
SHD4
SHD3
SHD2
SHD1
SHD0
SHA[0..12]
74LS157
E8
F3
E3
E2
SA0
SA1
SA2
SA3
SA4
SA5
SA6
SA7
SA8
SA9
SA10
SA11
SA12
MA0
MA1
MA2
MA3
MA4
MA5
MA6
MA7
MA8
MA9
MA10
MA11
MA12
MRD
SRD
D3
74HC245
D4
74HC245
buffers
A->B
MBD0
MBD1
MBD2
MBD3
MBD4
MBD5
MBD6
MBD7
SD0
SD1
SD2
SD3
SD4
SD5
SD6
SD7
SHD[0..7]
U3
Z80CPU
A3A
74LS107
SCK6MD
RESET
CLK
NMI
INT
M1
RFSH
WAIT
HALT
BUSRQ
BUSACK
RD
WR
MREQ
IORQ
SRD
SWR
SRFSH
SA[0..15]
SD[0..7]
SBSWD[0..5]
SRA[13..14]
H5
74HC174
SBNKSW
SRST
B06-13
PAL16L8'
B53-25.U3
27512
SROMCS
SSHRAMRQ
SBNKSW
DIPSND
RDINPUTS
PALEN
Pin 12: 0x0000..0x7FFF
or 0x8000..0x9FFF
Pin 13: 0xE000..0xEFFF
or 0xD000..0xDFFF
Pin 14: 0xA000..0xAFFF
Pin 15: 0xB000..0xBFFF
Pin 16: 0xC000..0xCFFF
Pin 17: 0xF000..0xFFFF
SROMCS = (SA14 & SA13 + SA15) & SMREQ & SRFSH
SSHRAMRQ = (SA13 & SA12 + SA13 & SA12) & SA15 & SA14 & SMREQ & SRFSH
SBNKSW = SA15 & SA14 & SA13 & SA12 & SWR & SMREQ & SRFSH
DIPSND = SA15 & SA14 & SA13 & SA12 & SMREQ & SRFSH
RDINPUTS = SA15 & SA14 & SA13 & SA12 & SMREQ & SRFSH
PALEN = SA15 & SA14 & SA13 & SA12 & SMREQ & SRFSH
SRA13 = SA15 & SBSWD0 + SA15 & SA13
SRA14 = SA15 & SBSWD1 + SA15 & SA14
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Sheet: /CPU/
File: cpu.kicad_sch
Title: The NewZealand Story – P0-043A
Size: A3
Date: 2024-04-07
KiCad E.D.A. 8.0.9
Rev:
Id: 3/7

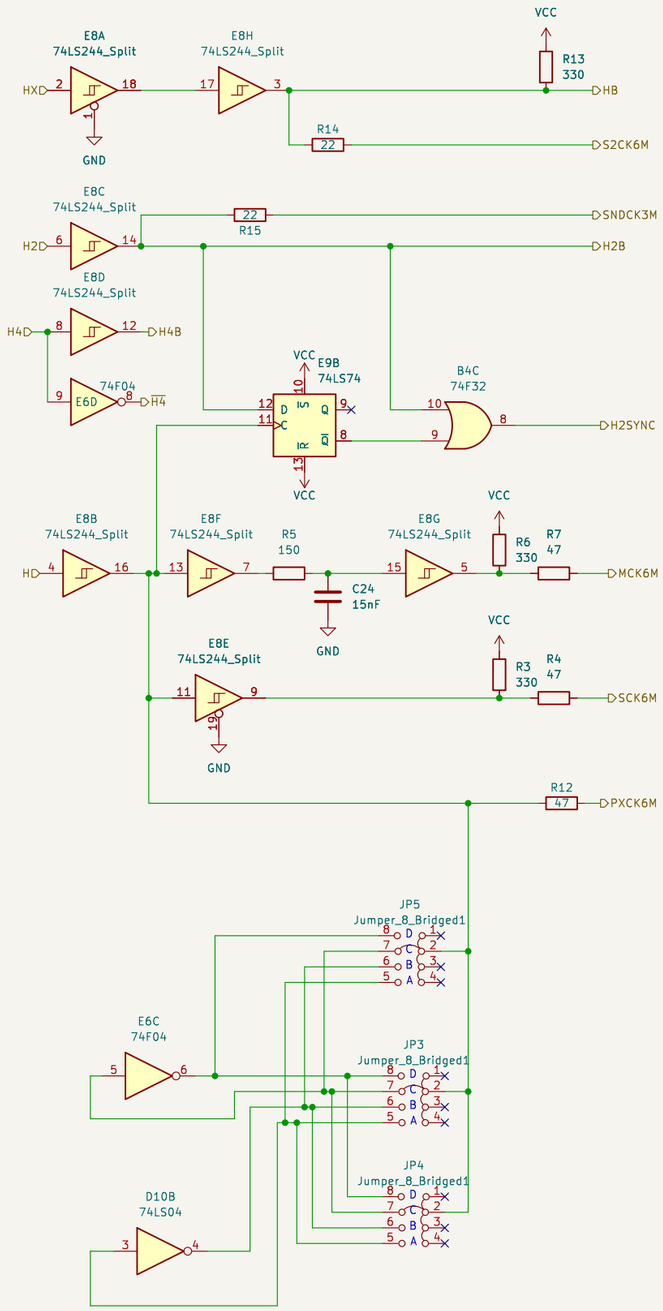
CC3
74HC245
MBD[0..7]
MBD7
MBD6
MBD5
MBD4
MBD3
MBD2
MBD1
MBD0
A0
A1
A2
A3
A4
A5
A6
A7
buffers
B0
B1
B2
B3
B4
B5
B6
B7
VD7
VD6
VD5
VD4
VD3
VD2
VD1
VD0
D5D
74LS32
H4B
OBJORVRAMEN
MRD
A->B
CE
D6
74HC374
D0
D1
D2
D3
D4
D5
D6
D7
O0
O1
O2
O3
O4
O5
O6
O7
VDB7
VDB6
VDB5
VDB4
VDB3
VDB2
VDB1
VDB0
3-state
flip flops
H2SYNC
H4
Cp
OE
CC6
OSC-12.0000MHz
VCC
GND
OUT
NC
R1
330
R2
22
VDB[0..7]
VA[0..12]
MRD
E6A
74F04
OBJRD
B4D
74F32
E6B
74F04
H8
74LS253
MA12
I0a
I1a
I2a
I3a
4-to-1 mux
OEa
Za
I0b
I1b
I2b
I3b
4-to-1 mux
OEb
Zb
A0
A1
A5
6264SL
VA12
VA11
VA10
VA9
VA8
VA7
VA6
VA5
VA4
VA3
VA2
VA1
VA0
A12
A11
A10
A9
A8
A7
A6
A5
A4
A3
A2
A1
A0
R/W
CS2
CS1
OE
D7
D6
D5
D4
D3
D2
D1
D0
OBJRAMCS
MA[0..11]
74HC245
MA5
MA4
MA3
MA2
MA1
MA0
VA5
VA4
VA3
VA2
VA1
VA0
VDB2
A2 and B2 use a modified symbol
with A inputs fixed as inputs
in order to avoid DRC errors.
D8
YM3906
CLK0
12M
VBLK
HBLK2
VSYNC
HBLK
DATAEN
HC8
HC7
HC6
HC5
HC4
HC3
HC2
HC1
HC0
CHA0
CHA1
CHA2
CHA4
GA0
GA1
GA2
GA4
H8B
H8R
SETA_X1-001
SPBANK2
SPBANK1
MA11
-{WDRES
A11
VA11
OBJRD
VA10
VA9
VA8
VA7
VA6
MAB11
VDB7
VDB6
VDB5
VDB4
VDB3
VDB2
VDB1
VDB0
H4
H2
H
HX
{HCNT}
CC8
YM3909
BGTRNSP
CLK
PAL4/TRSP
PAL3
PAL2
PAL1
PAL0
UK
DATAEN
HC8
HC7
HC6
HC5
HC4
HC3
HC2
HC1
HC0
GDA0
GDA1
GDA2
GDA3
GDA4
GDA5
GDA6
GDA7
GDB0
GDB1
GDB2
GDB3
GDB4
GDB5
GDB6
GDB7
GDC0
GDC1
GDC2
GDC3
GDC4
GDC5
GDC6
GDC7
GDD0
GDD1
GDD2
GDD3
GDD4
GDD5
GDD6
GDD7
SETA_X1-002
SA[0..9]
B11
X1-006
SRD
SWRD
PALEN
RD
WR
SD[0..7]
SD0
SD1
SD2
SD3
SD4
SD5
SD6
SD7
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
COL0
COL1
COL2
COL3
COL4
COL5
COL6
COL7
COL8
SA1
SA2
SA3
SA4
SA5
SA6
SA7
SA8
SA9
SA0
BB12
X1-007
RGB1
RGB2
RGB3
RGB4
RGB5
RGB6
RGB7
RGB8
RGB9
RGB10
RGB11
RGB12
RGB13
RGB14
RGB15
PXCLK
VBLK
HSYNC
CSYNC
PXCK6MD
R[1..6]
G[1..6]
B[1..6]
R0-012
RED
GREEN
BLUE
R8
47
SYNC
C25
150nF
EE10
74HC161
F10A
74HC14
counter
74HC273
flip flops
74HC174
hex flip flop
GROMEN
{GA}
From sub PCB connectors
To sub PCB connectors
GDD[0..7]
GDC[0..7]
GDB[0..7]
GDA[0..7]
VCC
GND
VSS

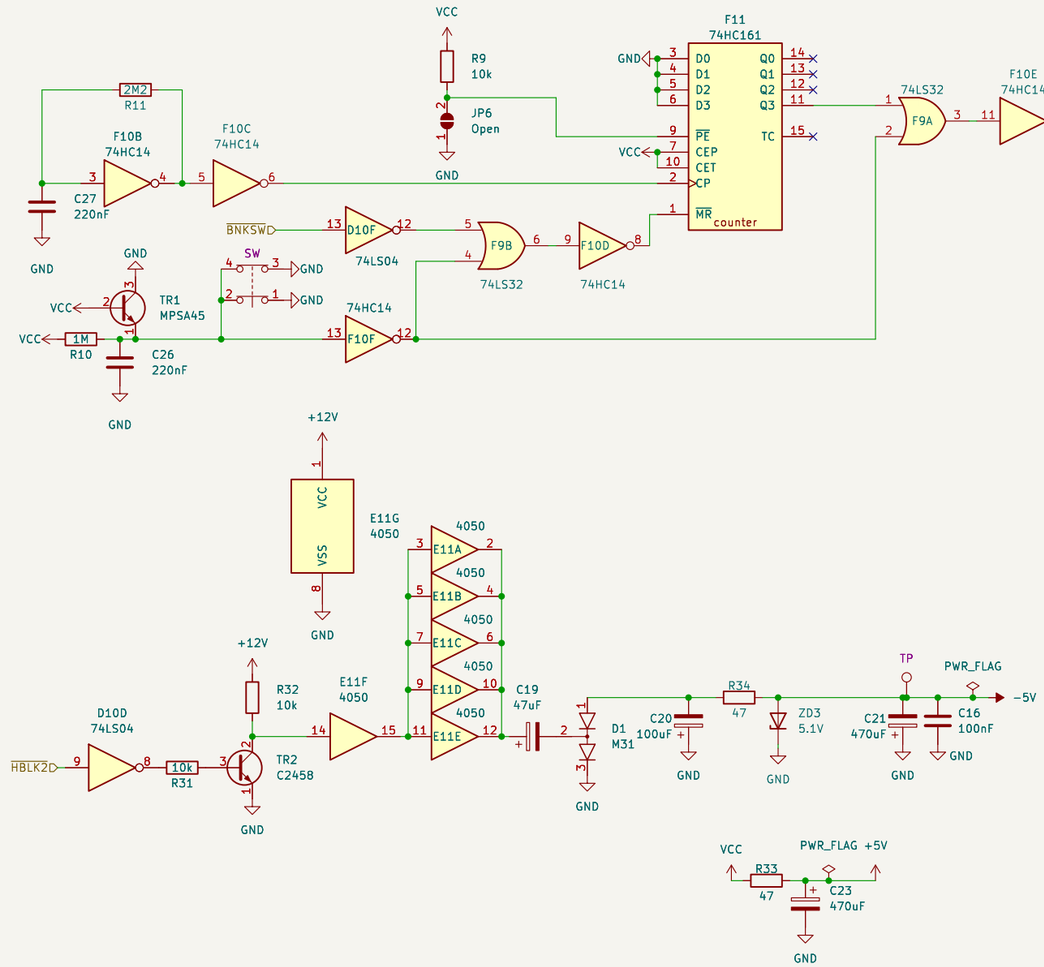
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Sheet: /Graphics/
File: graphics.kicad_sch

Title: The NewZealand Story – P0-043A

Size: A3	Date: 2024-04-07	Rev:
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Sheet: /Misc/

File: misc.kicad_sch

Title: The NewZealand Story – P0-043A

Size: A3	Date: 2024-04-07	Rev:
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RA3
2.2K x 8
1P_START 2
1P_BTN_3 3
1P_BTN_2 4
1P_BTN_1 5
1P_DOWN 6
1P_UP 7
1P_RIGHT 8
1P_LEFT 9
1 VCC

RA4
2.2K x 8
2P_START 2
2P_BTN_3 3
2P_BTN_2 4
2P_BTN_1 5
2P_DOWN 6
2P_UP 7
2P_RIGHT 8
2P_LEFT 9
1 VCC

RA5
2.2k x 4
TILT 2
SERVICE 3
COIN_2 4
COIN_1 5
1 VCC

JAMMA_CONN
PWR_FLAG
GND 1
PWR_FLAG 2
VCC 3
4
5
6
+12V
COIN_CNT_1 8
COIN_LOCK_1 9
SPEAKER+ 10
JAMMA_11D 11
REDD 12
BLUED 13
GND 14
15
COIN_1 16
1P_START 17
1P_UP 18
1P_DOWN 19
1P_LEFT 20
1P_RIGHT 21
1P_BTN_1 22
1P_BTN_2 23
1P_BTN_3 24
25
26
27
GND 28
COIN_CNT_1 LOCK_COIL_1 SPEAKER_(+) NOT_USED RED BLUE VIDEO_GND TEST COIN_1 1P_START 1P_UP 1P_DOWN 1P_LEFT 1P_RIGHT 1P_BTN_1 1P_BTN_2 1P_BTN_3 1P_BTN_4 1P_SELECT GND GND
COIN_CNT_2 LOCK_COIL_2 SPEAKER_(-) NOT_USED GREEN SYNC SERVICE TILT COIN_2 2P_START 2P_UP 2P_DOWN 2P_LEFT 2P_RIGHT 2P_BTN_1 2P_BTN_2 2P_BTN_3 2P_BTN_4 2P_SELECT GND GND
A GND
B GND
C VCC
D
E
F
PWR_FLAG C1 470uF +12V
ZD1 > 5.0V
GND GND
J COIN_CNT_2
K COIN_LOCK_2
L
M SPEAKER-
N GREEN
P SYNC
R SERVICE
S TILT
T COIN_2
U 2P_START
V 2P_UP
W 2P_DOWN
X 2P_LEFT
Y 2P_RIGHT
Z 2P_BTN_1
a 2P_BTN_2
b 2P_BTN_3
c
d
e
f GND

+12V
ZD2
> 12.0V
C4 220uF
GND GND

E12 X2-004
VCC 1
1P_START 2
1P_BTN_3 3
1P_BTN_2 4
1P_BTN_1 5
1P_DOWN 6
1P_UP 7
VCC 8
VCC1 VCC1 IN IN IN IN IN IN VCC2 VCC2
2k2 1k 2k2 1k
GND 16 OUT 15 OUT 14 OUT 13 OUT 12 OUT 11 OUT 10 GND 9
1P_START_B 1P_BTN_3_B 1P_BTN_2_B 1P_BTN_1_B 1P_DOWN_B 1P_UP_B

D12 X2-004
VCC 1
1P_RIGHT 2
1P_LEFT 3
2P_START 4
2P_BTN_3 5
2P_BTN_2 6
2P_BTN_1 7
VCC 8
VCC1 VCC1 IN IN IN IN IN IN VCC2 VCC2
2k2 1k 2k2 1k
GND 16 OUT 15 OUT 14 OUT 13 OUT 12 OUT 11 OUT 10 GND 9
1P_RIGHT_B 1P_LEFT_B 2P_START_B 2P_BTN_3_B 2P_BTN_2_B 2P_BTN_1_B

CC12 X2-004
VCC 1
2P_DOWN 2
2P_UP 3
2P_RIGHT 4
2P_LEFT 5
TILT 6
SERVICE 7
VCC 8
VCC1 VCC1 IN IN IN IN IN IN VCC2 VCC2
2k2 1k 2k2 1k
GND 16 OUT 15 OUT 14 OUT 13 OUT 12 OUT 11 OUT 10 GND 9
2P_DOWN_B 2P_UP_B 2P_RIGHT_B 2P_LEFT_B TILT_B SERVICE_B

B12 X2-004
VCC 1
COIN_2 2
COIN_1 3
4
5
6
7
VCC 8
VCC1 VCC1 IN IN IN IN IN IN VCC2 VCC2
2k2 1k 2k2 1k
GND 16 OUT 15 OUT 14 OUT 13 OUT 12 OUT 11 OUT 10 GND 9
COIN_2_B COIN_1_B

E10A
74LS139
GND
SA1 2
RDINPUTS 1
A1 A0 E
O0 O1 O2 O3
4 5 6 7
decoder

D11
X1-004
SA0 37
E3 41
E2
E1 40
SD1 35
SD0 34
SD5 33
SD4 32
IOD1 IOD2 IOD3 IOD4
IOC1 18 TILT_B
IOC2 19 SERVICE_B
IOC3 20 COIN_2_B
IOC4 21 COIN_1_B
1P_START_B 2
1P_BTN_3_B 3
1P_BTN_2_B 4
1P_BTN_1_B 5
1P_DOWN_B 6
1P_UP_B 7
1P_RIGHT_B 8
1P_LEFT_B 9
INA1 INA2 INA3 INA4 INA5 INA6 INA7 INA8
D7 51 SD7
D6 50 SD6
D5 49 SD5
D4 48 SD4
D3 47 SD3
D2 46 SD2
D1 45 SD1
D0 44 SD0
2P_START_B 10
2P_BTN_3_B 11
2P_BTN_2_B 12
2P_BTN_1_B 13
2P_DOWN_B 14
2P_UP_B 15
2P_RIGHT_B 16
2P_LEFT_B 17
INB1 INB2 INB3 INB4 INB5 INB6 INB7 INB8
COIN_LOCK_1_B 22
COIN_LOCK_2_B 23
COIN_CNT_2_B 24
COIN_CNT_1_B 25
IOA1 IOA2 IOA3 IOA4
IOB1 31 (SBSWD5)
IOB2 30 (SBSWD5)
IOB3 29 SBSWD3
IOB4 28 SBSWD2
38 UK
36 UK
43 UK
42 UK
39 UK
VCC 52
VCC 27
GND 26
1
GND
VCC
GND
SBSWD[2..5]

The I/O direction of each port in X1-004 is probably determined by connecting the pins 36,38,39, 42 and 43 to VCC and GND accordingly.

In tnzs IOD -> IOC, IOB -> IOA

E1	E2	E3	Active output
H	L	L	INA-> D, IOB -> IOA
H	L	H	INB -> D, IOB -> IOA
L	H	L	IOC -> IOD, IOB -> IOA
X	X	X	IOB -> IOA

All inputs are active low, when the outputs are not active they are in tri-state state. Except for IOA in this case.

SBSWD4 JP7 SBSWD5

F12
COIN_LOCK_2_B 3
COIN_CNT_2_B 6
COIN_CNT_1_B 11
COIN_LOCK_1_B 14
I1 I2 I3 I4
O1 2 COIN_LOCK_2
O2 7 COIN_CNT_2
O3 9 COIN_CNT_1
O4 16 COIN_LOCK_1
TD62064
+12V 1
8
10
15
COM GND
COM GND
NC GND
NC GND
4 5 12 13
GND

RA2
8 x 10k
VCC 1
SWA1 2
SWA2 3
SWA3 4
SWA4 5
SWA5 6
SWA6 7
SWA7 8
SWA8 9

SWA
SW_DIP_x08
GND 16
15
14
13
12
11
10
9
SWA1 1
SWA2 2
SWA3 3
SWA4 4
SWA5 5
SWA6 6
SWA7 7
SWA8 8

RA1
8 x 10k
VCC 1
SWB1 2
SWB2 3
SWB3 4
SWB4 5
SWB5 6
SWB6 7
SWB7 8
SWB8 9

SWB
SW_DIP_x08
GND 16
15
14
13
12
11
10
9
SWB1 1
SWB2 2
SWB3 3
SWB4 4
SWB5 5
SWB6 6
SWB7 7
SWB8 8

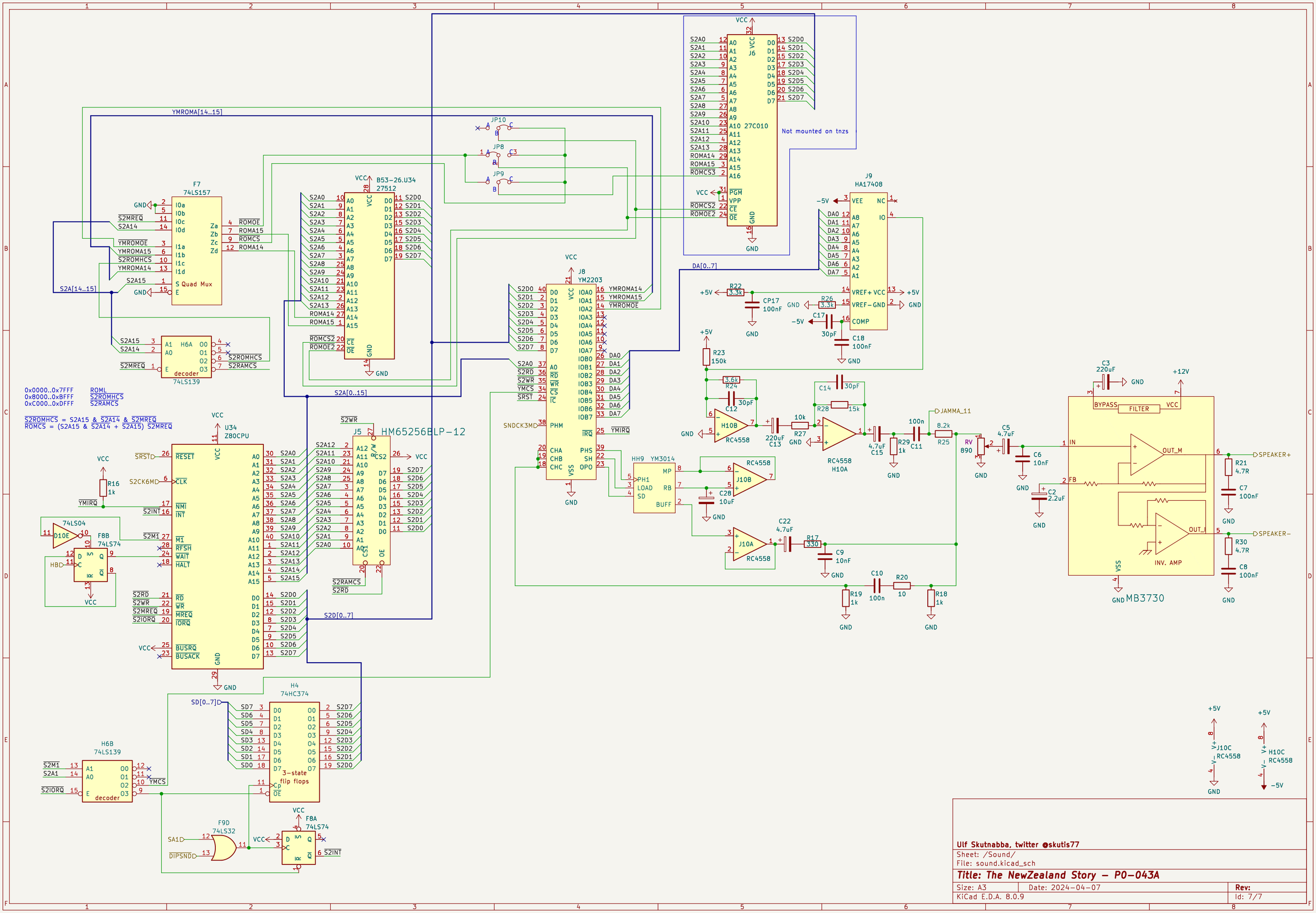
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SWB[1..8]

H3
74HC257
SWA1 2
SWB1 3
SWA2 5
SWB2 6
SWA4 14
SWB4 13
SWA3 11
SWB3 10
SA0 1
15
I0a I1a I0b I1b I0c I1c I0d I1d S OE
Za 4 SD0
Zb 7 SD1
Zc 12 SD3
Zd 9 SD2

H2
74HC257
SWA5 2
SWB5 3
SWA6 5
SWB6 6
SWA8 14
SWB8 13
SWA7 11
SWB7 10
SA0 1
15
I0a I1a I0b I1b I0c I1c I0d I1d S OE
Za 4 SD4
Zb 7 SD5
Zc 12 SD7
Zd 9 SD6

F9C
74LS32
SA2 9
DIPSND 10
8
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SD[0..7]
SD[0..7]

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Size: A3 | Date: 2024-04-07 | Rev:
KiCad E.D.A. 8.0.9 | Id: 6/7



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